

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1001 - Relief valve for C-101		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 1	
GENERAL					
1	Tag / quantity	PSV-1001 / 4	2	Protected equipment	C-101
3	Governing case	Reflux failure / blocked OH (total OH vapour)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	8T10 (inlet x orifice x outlet)	8	Inlet flange / facing	8" 150# RF
9	Outlet flange / facing	10" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	54,055 kg/h
12	Molecular weight	97.9	13	Relieving temperature	154 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	2.4 barg
16	Overpressure / accumulation	10 %	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	51,101 mm2
20	Selected orifice / effective area	T / 16,774 mm2 each	21	Utilisation	76 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1002 - Relief valve for D-101A		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 2	
GENERAL					
1	Tag / quantity	PSV-1002 / 1	2	Protected equipment	D-101A
3	Governing case	Fire (liquid full)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	4M6 (inlet x orifice x outlet)	8	Inlet flange / facing	4" 300# RF
9	Outlet flange / facing	6" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	50,292 kg/h
12	Molecular weight	120.0	13	Relieving temperature	200 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	15.5 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	2,286 mm2
20	Selected orifice / effective area	M / 2,323 mm2 each	21	Utilisation	98 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1003 - Relief valve for D-101B		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 3	
GENERAL					
1	Tag / quantity	PSV-1003 / 1	2	Protected equipment	D-101B
3	Governing case	Fire (liquid full)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	4M6 (inlet x orifice x outlet)	8	Inlet flange / facing	4" 300# RF
9	Outlet flange / facing	6" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	50,292 kg/h
12	Molecular weight	120.0	13	Relieving temperature	200 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	15.5 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	2,286 mm2
20	Selected orifice / effective area	M / 2,323 mm2 each	21	Utilisation	98 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1004 - Relief valve for D-102		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 4	
GENERAL					
1	Tag / quantity	PSV-1004 / 1	2	Protected equipment	D-102
3	Governing case	Fire	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	4P6 (inlet x orifice x outlet)	8	Inlet flange / facing	4" 150# RF
9	Outlet flange / facing	6" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	12,890 kg/h
12	Molecular weight	90.0	13	Relieving temperature	120 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	2.4 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	3,049 mm2
20	Selected orifice / effective area	P / 4,116 mm2 each	21	Utilisation	74 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1005 - Relief valve for C-105		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 5	
GENERAL					
1	Tag / quantity	PSV-1005 / 1	2	Protected equipment	C-105
3	Governing case	Reflux failure (reboiler duty)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	6R8 (inlet x orifice x outlet)	8	Inlet flange / facing	6" 150# RF
9	Outlet flange / facing	8" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	102,070 kg/h
12	Molecular weight	54.2	13	Relieving temperature	94 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	13.0 barg
16	Overpressure / accumulation	10 %	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	7,175 mm2
20	Selected orifice / effective area	R / 10,323 mm2 each	21	Utilisation	70 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1006 - Relief valve for D-105		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 6	
GENERAL					
1	Tag / quantity	PSV-1006 / 1	2	Protected equipment	D-105
3	Governing case	Fire	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	1.5H3 (inlet x orifice x outlet)	8	Inlet flange / facing	1.5" 150# RF
9	Outlet flange / facing	3" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	5,902 kg/h
12	Molecular weight	50.0	13	Relieving temperature	70 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	15.2 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	360 mm2
20	Selected orifice / effective area	H / 506 mm2 each	21	Utilisation	71 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1007 - Relief valve for C-106		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 7	
GENERAL					
1	Tag / quantity	PSV-1007 / 1	2	Protected equipment	C-106
3	Governing case	Reflux failure (reboiler duty)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	8T10 (inlet x orifice x outlet)	8	Inlet flange / facing	8" 150# RF
9	Outlet flange / facing	10" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	76,867 kg/h
12	Molecular weight	81.0	13	Relieving temperature	99 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	3.5 barg
16	Overpressure / accumulation	10 %	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	14,019 mm2
20	Selected orifice / effective area	T / 16,774 mm2 each	21	Utilisation	84 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1008 - Relief valve for E-116		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 8	
GENERAL					
1	Tag / quantity	PSV-1008 / 1	2	Protected equipment	E-116
3	Governing case	Tube rupture (HP steam into naphtha)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	4P6 (inlet x orifice x outlet)	8	Inlet flange / facing	4" 300# RF
9	Outlet flange / facing	6" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Steam / vapour	11	Required capacity (per valve)	25,000 kg/h
12	Molecular weight	18.0	13	Relieving temperature	260 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	13.0 barg
16	Overpressure / accumulation	10 %	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	3,673 mm2
20	Selected orifice / effective area	P / 4,116 mm2 each	21	Utilisation	89 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1009 - Relief valve for C-102/103/104		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 9	
GENERAL					
1	Tag / quantity	PSV-1009 / 1	2	Protected equipment	C-102/103/104
3	Governing case	Fire (side strippers)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	4N6 (inlet x orifice x outlet)	8	Inlet flange / facing	4" 150# RF
9	Outlet flange / facing	6" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	15,948 kg/h
12	Molecular weight	180.0	13	Relieving temperature	300 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	3.5 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	2,419 mm2
20	Selected orifice / effective area	N / 2,800 mm2 each	21	Utilisation	86 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-1010 - Relief valve for P-101 disch.		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 10	
GENERAL					
1	Tag / quantity	PSV-1010 / 1	2	Protected equipment	P-101 disch.
3	Governing case	Blocked outlet / thermal	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Conventional	6	Discharge to	Closed drain
7	Size designation (API 526)	1E2 (inlet x orifice x outlet)	8	Inlet flange / facing	1" 300# RF
9	Outlet flange / facing	2" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	5,000 kg/h
12	Molecular weight	150.0	13	Relieving temperature	60 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	30.0 barg
16	Overpressure / accumulation	10 %	17	Back pressure	atm
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	91 mm2
20	Selected orifice / effective area	E / 126 mm2 each	21	Utilisation	72 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	-
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125..137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-2001 - Relief valve for C-201		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 11	
GENERAL					
1	Tag / quantity	PSV-2001 / 1	2	Protected equipment	C-201
3	Governing case	Loss of vacuum / fire (blocked outlet to ejectors)	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	6R8 (inlet x orifice x outlet)	8	Inlet flange / facing	6" 150# RF
9	Outlet flange / facing	8" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	32,789 kg/h
12	Molecular weight	60.0	13	Relieving temperature	400 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	3.5 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	9,337 mm2
20	Selected orifice / effective area	R / 10,323 mm2 each	21	Utilisation	90 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PRESSURE RELIEF VALVE DATASHEET API 526 / API 520 PSV-2002 - Relief valve for D-201		DOC. No. CFU-000-ME-DS-007 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 12	
GENERAL					
1	Tag / quantity	PSV-2002 / 1	2	Protected equipment	D-201
3	Governing case	Fire	4	Valve type	Spring-loaded, full-nozzle, high-lift
5	Bonnet	Balanced bellows	6	Discharge to	Flare
7	Size designation (API 526)	3K4 (inlet x orifice x outlet)	8	Inlet flange / facing	3" 150# RF
9	Outlet flange / facing	4" 150# RF			
SERVICE CONDITIONS & SIZING (API 520)					
10	Fluid / state	Hydrocarbon vapour	11	Required capacity (per valve)	6,324 kg/h
12	Molecular weight	100.0	13	Relieving temperature	120 C
14	Cp/Cv (k) / Z	1.10 / 1.0	15	Set pressure	3.5 barg
16	Overpressure / accumulation	21 % (fire)	17	Back pressure	0.5 barg superimposed (var.) + 10 % built-up
18	Kd / Kb / Kc	0.975 / 1.0 / 1.0	19	Required orifice area (total)	1,066 mm2
20	Selected orifice / effective area	K / 1,186 mm2 each	21	Utilisation	90 %
22	Inlet loss limit	< 3 % of set (API 520-II)			
MATERIALS & ACCESSORIES					
23	Body / bonnet	A216 WCC	24	Nozzle / disc	316 SS
25	Spring	CS, aluminised (W < 230 C) / tungsten (> 230 C)	26	Bellows	Inconel 625
27	Trim	316 SS (Stellited seats)	28	Lifting lever / test gag	No / Yes
29	NACE	MR0103 (sour)	30	Code	ASME VIII-1 UG-125...137, API 520/521/526/527
NOTES					
1. API 526 datasheet layout. Sizing per process (data/psv.json); vendor to confirm certified capacity (API 527 seat tightness). 2. Multiple valves: staggered set pressures (+5 %) where count > 1.					